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PAPER

An Unsupervised Opinion Mining Approach for Japanese Weblog Reputation Information Using an Improved SO-PMI Algorithm

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SUMMARY In this paper, we propose an improved SO-PMI (Semantic Orientation Using Pointwise Mutual Information) algorithm, for use in Japanese Weblog Opinion Mining. SO-PMI is an unsupervised approach proposed by Turney that has been shown to work well for English. When this algorithm was translated into Japanese naively, most phrases, whether positive or negative in meaning, received a negative SO. For dealing with this slanting phenomenon, we propose three improvements: to expand the reference words to sets of words, to introduce a balancing factor and to detect neutral expressions. In our experiments, the proposed improvements obtained a well-balanced result: both positive and negative accuracy exceeded 62%, when evaluated on 1,200 opinion sentences sampled from three different domains (reviews of Electronic Products, Cars and Travels from Kakaku.com). In a comparative experiment on the same corpus, a supervised approach (SA-Demo) achieved a very similar accuracy to our method. This shows that our proposed approach effectively adapted SO-PMI for Japanese, and it also shows the generality of SO-PMI.

key words: SO-PMI, opinion mining, Weblog reputation information, sentiment analysis, unsupervised learning, supervised learning

1. Introduction

Recently, more and more websites add information in the form of personal opinions to the Web, e.g. customer reviews of products, forums, discussion groups, and blogs. Here, we use the term Weblog for these sites. This type of information is often useful. However, we have to deal with an enormous amount of unstructured and/or semi-structured data. These data are subjective, in free format and mostly textual. Using them is thus difficult and time consuming. Therefore, the ability to more effectively mine Weblog opinions automatically has attracted more and more attention [1]–[3].

Weblog opinion information exists in various forms across different websites. It can appear on commercial product sites with complementary reviews (like Amazon), professional review sites (such as review.japan.zdnet.com), consumer opinion sites on broad topics and products (such as kakaku.com), news or magazines with feature reviews (www3.keizaireport.com), personal Weblog sites (such as blog.goo.ne.jp) or even on bulletin boards and Usenet groups that archive less formal reviews. These reviews are usually based on opinions expressed in very unstructured and free formats, and mix positive, negative and non-opinion information (factual descriptions) [4]–[7].

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Text mining is a cross between Information Retrieval (IR), Data Mining (DM), Artificial Intelligence (AI), and Natural Language Processing (NLP). It is also known as intelligent text analysis, text data mining or Knowledge Discovery in Text (KDT), and generally refers to the process of extracting information and knowledge, facts and meaningful relationships, from text documents. Since most company data and Web sources are stored as text, the purpose of text mining is to discover important and valuable information from these massive unstructured text data. Opinion mining is a subdiscipline of text mining. It aims to analyze the reputations of various products automatically by classifying them into positive or negative meaning from the "subjective" terms contained in the document. It can play an important role in providing a speedy search and analysis result to both consumers and manufacturers [8].

Text mining methods can be divided into two types: supervised and unsupervised approaches. Both of these approaches have their pros and cons. Supervised approaches are likely to provide more accurate classification results but require a training corpus, and require retraining if they are to be applied in a new domain. Thus, they are costly (labor, money) to use because of the need for large training resources. Unsupervised approaches on the other hand do not require any manual annotation, but the results are usually worse than for supervised methods. These approaches use statistics from unlabeled text to measure how far towards positive or negative a word is inclined, and do not require a prior training corpus. Nowadays, Web resource, usually used as a free corpus, is rapidly expanding. Therefore, how to use this open resource more effectively by unsupervised approaches has seen increasing attention over the last five years [3].

Turney [9] has presented an unsupervised opinion mining algorithm called SO-PMI (Semantic Orientation Using Pointwise Mutual Information). The main use of SO-PMI is to estimate the semantic orientation (i.e. positive or negative) of a phrase by measuring the hits for pairs of words or phrases returned from a search engine, based on the mutual information theory. This approach has previously been successfully used on English. The average accuracy was 74% when evaluated on 410 reviews from Epinions*.

However, according to our preliminary experiment, naively translating Turney's original idea into Japanese gave a very slanted result, with a *Negative Accuracy* of 94.3%

*http://www.epinions.com/

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and a *Positive Accuracy* of only 14.5%. We found that the balance between the positive and negative sides is influenced greatly by the number of page hits of the reference words, since a search engine is used. Therefore, we introduced reference sets that have higher and more stable frequencies, and a balancing factor adjusting for the difference in occurrence between positive and negative words. We also added several threshold rules to detect neutral expressions. The proposed approach is evaluated on three domains (Electronic Products, Cars and Travels). For each domain a dataset with 200 positive and 200 negative Japanese opinion sentences was used and our method was compared to a supervised approach. The results showed that the accuracy of the improved SO-PMI algorithm not only yielded a well-balanced result but also compares well with the traditional supervised approach. It is also shown that the proposed approach does not depend on a special domain but can be used in opinion mining of various domains.

The remainder of this paper is organized as follows: Section 2 describes the original SO-PMI Algorithm in detail and the problems with a naive translation of SO-PMI to Japanese. In Sect. 3, we present the three improvements to SO-PMI for Japanese. Then Sect. 4 describes the data for the experiments, the balancing factor, an evaluation of each improvement and a comparative experiment. As an example application of the SO-PMI algorithm, a simple opinion mining system is outlined in Sect. 5. Finally, related work and conclusions are presented in Sects. 6 and 7.

2. The SO-PMI Algorithm

2.1 Basic Theory of SO-PMI

The SO-PMI algorithm [9] is used to estimate the semantic orientation (SO) of a phrase by measuring the similarity of pairs of words or phrases using the following formula:

$$PMI(word_1, word_2) = log_2\left[\frac{p(word_1 \& word_2)}{p(word_1)\ p(word_2)}\right] \quad (1)$$

Assumption : *Words co-occurring with positive (negative)words tend to be positive(negative) too.*

$$SO(phrase) = PMI(phrase, \text{“}excellent\text{”}) - PMI(phrase, \text{“}poor\text{”}) \quad (2)$$

The reason for choosing the reference words "excellent" and "poor" is that in five star review rating systems it is common to define one star as "poor" and five stars as "excellent". SO is positive when a phrase is more strongly associated with "excellent" and negative when a phrase is more strongly associated with "poor". Let hits(query) be the number of hits returned when using a search engine, the following estimate of SO can be derived from Formula (1) and (2) with some minor algebraic manipulation.

< noun + auxiliary word + adj / verb / noun >
< adj + noun / undefined / verb >
< noun + verb >
< noun + symbol + adj / verb / noun >
Except the above < adj / verb >

Fig. 1 Opinion phrase templates.

$$SO(phrase) = log_2[A]$$
$$A = \frac{hits(phrase\ NEAR\ \text{“}excellent\text{”})\ hits(\text{“}poor\text{”})}{hits(phrase\ NEAR\ \text{“}poor\text{”})\ hits(\text{“}excellent\text{”})} \quad (3)$$

Turney used the AltaVista† search engine because it has a *NEAR* operator. This operator constrains the search to documents that contain the words within ten words of one another, in either order. Turney's previous work has shown that *NEAR* performs better than *AND* when measuring the strength of semantic association between words.

2.2 Naive Translation of SO-PMI for Japanese

For using SO-PMI in Japanese opinion mining, opinion phrases are first extracted according to opinion templates. The Weblog sentences are analyzed using a dependency structure analyzer for Japanese, CaboCha [10]. Then opinion expressions are extracted using word POS (part of speech) templates. Based on our analysis of opinions in Japanese reviews [11], [12] and the results of related work [13], [14], an opinion generally consists of two parts: the object name and opinion phrases. The POS of the object name is generally noun or undefined. The POS of the opinion phrase is generally adjective, verb or noun. Therefore, we designed the templates shown as Fig. 1 to collect opinion phrases.

Then, following Turney's original idea, we naively translated the SO formula to the one shown in Formula (4) for Japanese.

$$SO(phrase) = log_2[B]$$
$$B = \frac{hits(phrase\ AND\ \text{“よい”})\ hits(\text{“悪い”})}{hits(phrase\ AND\ \text{“悪い”})\ hits(\text{“よい”})} \quad (4)$$

We used the Google search engine†† to get the hits(query) even though Google does not have a *NEAR* operator. The AltaVista *NEAR* operator does not work well for Japanese and Google indexes more pages than AltaVista. Thus, we used Google and replaced the *NEAR* operator with the *AND* operator in the SO formula. "よい (good)" and "悪い (bad)" were selected because they are the most commonly used evaluation expressions in Japanese.

For testing the performance of this trial, we used the Common Corpus which is described in detail in 4.1.1. It has 1,000 positive and 1,000 negative sentences which were collected randomly from the Web. The results were

†http://www.altavista.com/
††http://www.google.co.jp/

very slanted (*Positive Accuracy*: 14.5%, *Negative Accuracy*: 94.3%). Many phrases, whether positive or negative in meaning, still received a negative SO. Some possible causes could be that "いい (good)" has more hits than "悪い (bad)", as shown in Table 2, and that the *AND* operator is less useful than the *NEAR* operator. We also tried to use the "*" wildcard and "*OR*" Google operator to simulate the *NEAR* operator. However, this made the queries very long, and Google search engine may not handle very complex queries as expected (some parts of the query return NULL). Thus, in the end we used the *AND* operator in this paper.

3. Proposed Method

We consider that the slanted results were influenced by characteristics of Japanese and the Google search engine. To deal with this problem, we evaluated the following three approaches - Reference Sets, Balance Factor and Neutral Expression Detection, to improve the SO-PMI algorithm for Japanese.

3.1 Reference Sets

In Japanese, there are many expressions being used when people evaluate something. For example, "いい (good)", "良い (good)", "満足 (satisfaction)" , "すばらしい (excellent)" are usually used when someone wants to convey a positive opinion. There are also no five star rating expressions as "excellent" and "poor" which are used widely in English. In Japanese, people often use five star marks or the numbers 1 to 5 instead of the review rating words. Hence, we selected two reference sets: "*p_set*" and "*n_set*" including the most common positive and negative words used in descriptions to substitute for the two single reference words. These are shown in Table 1.

"*p_set*" is a set of common positive words. "*n_set*" is a set of common negative words. The hit counts of these words from Google search engine is shown in Table 1 (All data from 2007/10/03). The "Hits" column shows the hit page counts for each word. The total number of Web pages is 1,010,000,000, calculated by taking the hits of "は (WA)". WA is a subject indicator particle and almost every Japanese page includes this particle. The "F" column of this table is the frequency in % of the total Web pages that contain the word. The bold and italic rows indicate the data used in the Reference Words/Sets Test (Table 2).

Table 1 Frequencies of *p_set*/*n_set* words on the Web.

p_set	Hits (K)	F(%)	*n_set*	Hits (K)	F(%)
いい(good)	***265,000***	***26.24***	***不良(bad)***	***147,000***	***14.55***
好き(like)	169,000	16.73	***悪い(bad)***	***74,000***	***7.33***
良い(good)	159,000	15.74	不安(worry)	60,600	6.00
魅力(charm)	125,000	12.38	難しい(hard)	55,600	5.50
よい(good)	77,400	7.66	嫌(dislike)	48,700	4.82
大好き(favorite)	77,300	7.65	嫌い(dislike)	27,700	2.74
良く(good)	77,100	7.63	不快(dissatisfaction)	22,200	2.20
楽しい(delightful)	76,000	7.52	***だめ(useless)***	***20,900***	***2.07***
満足(satisfaction)	***65,900***	***6.52***	辛い(painful)	19,600	1.94
面白い(interesting)	57,900	5.73	不具合 (fault)	18,400	1.82
嬉しい(happy)	53,500	5.30	***不満(dissatisfaction)***	***18,200***	***1.80***
素敵(lovely)	52,400	5.19	最悪(worst)	16,100	1.59
うれしい(happy)	36,400	3.60	いまいち(not good)	11,000	1.09
おもしろい(interesting)	21,600	2.14	まずい(bad)	7,690	0.76
すばらしい(wonderful)	***21,200***	***2.10***	欠点(fault)	6,840	0.68

- Selection Criteria

  The selection criteria of the "*p_set*" and "*n_set*" reference sets is based on the following steps.

  1. Use the common expressions (not limited to a special domain) from the evaluation expression dictionary made by Kobayashi et al. [13].
  2. Use words which have high frequency both on the Web and in the collected Weblog corpus (4.1.1).
  3. Use words which have a stable distribution over time on the Web. Stable distributions lead to more reliable results. For checking the distributions of words in the reference sets, we observed the frequencies of the "*p_set*" words (the upper part in Fig. 2) and the "*n_set*" words (the lower part in Fig. 2) on the Web, from 2006.07 to 2007.03 once every two months. The figure shows the words that have a comparatively steady frequency (except for the data from 2006.11). Note that words that had an unstable frequency were omitted.

- Reference Words/Sets Test

  Besides "いい (good)" and "悪い (bad)" [No.1], we also used other reference word pairs like "すばらしい (excellent)" and "不良 (poor)" [No.2], "すばらしい (excellent)" and "だめ (useless)" [No.3], "満足 (satisfaction)" and "不満 (dissatisfaction)" [No.4], along with the proposed reference sets ("*p_set*" and "*n_set*" [No.5]) and tested them on the Common Corpus. The [No.1] and [No.4] pairs were chosen because they are common evaluation expressions. [No.2] and [No.3] are frequently used to translate the English words "excellent" and "poor". We obtained the results shown in Table 2. These results show that all of them are slanted, either to the positive or negative. In Table 2, it can be noted that when hits of the negative reference words (sets) is higher than for the positive, the results are slanted towards *Positive Accuracy*, and vice versa. Pair [No.3] which has the most similar hits obtained a bal-

Table 2 Reference words/sets test.

No	Reference Words (Sets) Set-Value	Hits(K)	PA(%)	NA(%)	AA(%)
1	いい(good) 悪い(bad)	265,000 74,000	14.5	94.3	54.4
2	すばらしい(excellent) 不良(poor)	21,400 147,000	96.8	11.4	54.1
3	すばらしい(excellent) だめ(useless)	21,400 20,900	31.9	57.2	44.6
4	満足(satisfaction) 不満(dissatisfaction)	65,900 18,200	8.6	92.2	50.4
5	*p_set* [いい(good), 好き(like), ・・・・・・] *n_set* [不良(poor), 悪い(bad), ・・・・・・]	287,000 141,000	16.2	95	55.6

PA: Positive Accuracy NA: Negative Accuracy AA: The average of PA and NA

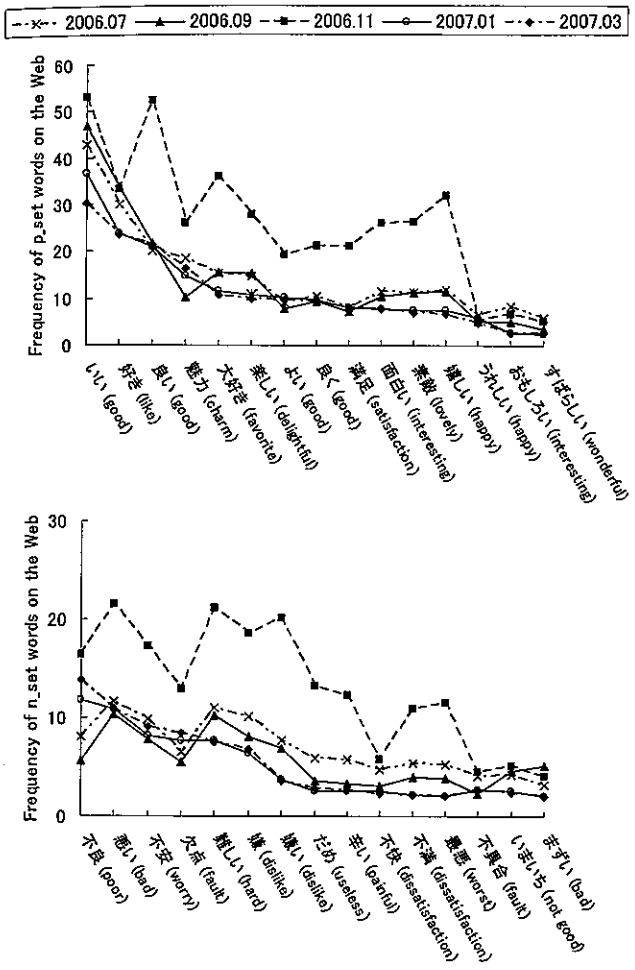


Fig. 2 Frequency over time for *p_set/n_set* words.

ance between positive and negative accuracy, but the lowest average accuracy (44.6%). Probably because the number of page hits is balanced but low. Moreover, the reference sets ("*p_set*" and "*n_set*") obtained the most hits and the highest average accuracy (55.6%).

Since the reference sets "*p_set*" and "*n_set*" have more hits (which means better coverage), more stable frequency and achieved more accurate results, as a first improvement of SO-PMI we replaced the reference words "いい (good)" and "悪い (bad)" with these two reference sets. Formula (4) was thus modified to (5). The *hits(P AND p_set)* was calculated by *hits(phrase AND ("いい" OR "好き" OR "良い" OR ...))*.

$$SO(P) = log_2\left[\frac{hits(P\ AND\ p_set)\ hits(n_set)}{hits(P\ AND\ n_set)\ hits(p_set)}\right] \quad (5)$$

P : *phrase*

p_set : *a set of common positive words*

n_set : *a set of common negative words*

3.2 Balancing Factor

In the experiments above, we obtained heavily slanted results, even for the proposed reference sets. We consider that the large difference in page hits between the positive and negative reference words/sets is the main cause for this phenomenon. To mitigate this problem, we introduced a balancing factor to adjust the balance between the positive and negative sides.

Formula (5) can be rewritten as:

$$SO(P) = log_2\left[\frac{hits(P\ AND\ p_set)}{hits(P\ AND\ n_set)}\right] + log_2\left[\frac{hits(n_set)}{hits(p_set)}\right] \quad (6)$$

In Formula (6), $log_2\left[\frac{hits(n_set)}{hits(p_set)}\right]$ is independent of the phrase P. For convenience of notation, let $\phi = log_2\left[\frac{hits(n_set)}{hits(p_set)}\right]$. Thus, Formula (6) becomes:

$$SO(P) = log_2\left[\frac{hits(P\ AND\ p_set)}{hits(P\ AND\ n_set)}\right] + \phi \quad (7)$$

This formula shows that the $SO(P)$ function consists of a *log* function and a constant ϕ. The curve of the $SO(P)$ function is translated by ϕ. As mentioned for the preliminary experiment in Table 2, the SO values of phrases are slanted, towards positive or negative meaning depending on the reference words/sets used. Note that the slanting is influence chiefly by ϕ. Therefore, we introduce a balancing factor to adjust the balance between positive and negative SO values based on ϕ. Accordingly we propose Formula (8):

$$SO(P) = log_2\left[\frac{hits(P\ AND\ p_set)}{hits(P\ AND\ n_set)}\right] + \phi + \lambda \cdot \phi \quad (8)$$

The purpose of using a weight λ is that the results will be slanted towards the opposite direction when subtracting a full ϕ. Thus the interesting range of λ is $[-1, 0]$. For convenience of notation, let "$\lambda \cdot \phi, \lambda \in [-1, 0]$" be rewritten as "$\alpha \cdot (-1)\phi, \alpha \in [0, 1]$". Thus, Formula (8) can be rewritten as Formula (9). we then get Formula (10):

$$SO(P) = log_2\left[\frac{hits(P\ AND\ p_set)}{hits(P\ AND\ n_set)}\right] + log_2\left[\frac{hits(n_set)}{hits(p_set)}\right] + \alpha \cdot log_2\left[\frac{hits(n_set)}{hits(p_set)}\right]^{-1} \quad (9)$$

$$SO(P) = log_2\left[\frac{hits(P\ AND\ p_set)\ hits(n_set)}{hits(P\ AND\ n_set)\ hits(p_set)}\right] + \alpha \cdot log_2\left[\frac{hits(p_set)}{hits(n_set)}\right] (P : phrase, \alpha \in [0, 1]) \quad (10)$$

We call "$\alpha \cdot log_2\left[\frac{hits(p_set)}{hits(n_set)}\right]$" a balancing factor. We find an appropriate α in later experiments using the *Harmonic*

Mean of *Positive Accuracy* and *Negative Accuracy*, as described in detail in 4.3. Using Formula (10), we achieved a well balanced *Positive Accuracy* and *Negative Accuracy* in the comparative experiment (4.6). Therefore, it is clear that the proposed balancing factor is effective.

3.3 Neutral Expression Detection

From the preliminary trials, we also observed that many neutral phrases received positive or negative SO. Neutral phrases mean phrases that do not include polarity, i.e. either positive or negative opinion. Therefore, as the third improvement for SO-PMI, we added detection of neutral expressions. The idea is that if a phrase is strongly or faintly associated with both "*p_set*" and "*n_set*", it is considered a neutral phrase. This means that the phrase has an ambiguous connection with both "*p_set*" and "*n_set*". We use the following rules (Fig. 3) to separate neutral phrases from positive/negative phrases. The threshold values $\mathbf{t_a}$, $\mathbf{t_b}$ and $\mathbf{t_c}$ are obtained using a small, hand-labeled corpus. The $\mathbf{t_a}$ and $\mathbf{t_b}$ are constants. The value of $\mathbf{t_c}$ is dependent on the hit counts of the phrase and the reference sets (min [(*hits*(*phrase AND p_set*), *hits*(*phrase AND n_set*)]).

Table 3 shows several sample SO values of this detection. Cases No.1-4 correspond to 1-4 of Fig. 3 respectively. Cases No.1-3 are examples of correct neutral expression detection. In Case No.4, "ザラザラ (roughly)" was not extracted as an opinion phrase by the opinion templates (Fig. 1) because it is an adverb. Thus it only received the default SO value 0. How to handle adverbs will be a future focus of our studies. Case No.5 is an example of misclassification. It shows that the positive phrase "信用度　向上する (improves reliability)" has an ambiguous connection with both "*p_set*" and "*n_set*" based on the hits. Phrases such as cases 4 and 5 accounted for under 30 percent of the detected neutral phrases. Therefore, we considered that this detection is effective in improving the SO-PMI algorithm.

1. $hits(phrase\ AND\ p_set) > t_a$ *AND* $hits(phrase\ AND\ n_set) > t_a$
2. $hits(phrase\ AND\ p_set) < t_b$ *AND* $hits(phrase\ AND\ n_set) < t_b$
3. $|\ hits(phrase\ AND\ p_set) - hits(phrase\ AND\ n_set)\ | < t_c$
4. $SO(phrase) = 0$

Fig. 3 Rules for detecting neutral expression.

Table 3 Examples of neutral expression detection.

Case No.	Phrase	hits (*phrase* AND *p_set*)	hits (*phrase* AND *n_set*)	SO	Detection Rule
1	なった (become)	10,300,000	10,100,000	-0.24	1
2	家の無線LAN 接続出来て (Wireless LAN of the house, can be connected)	4	3	0.15	2
3	PS2版日本語 変換効率 (PS2 edition Japanese, conversion efficiency)	20,300	20,000	-0.35	3
4	画面がザラザラ (The screen is roughly)	-	-	0	4
5	信用度 向上する (improves reliability)	197	193	-0.23	3

SO was calculated by Formula (10).

4. Performance Evaluation

4.1 Two Corpora for Experiments

In this paper, we used the Common Corpus for the preliminary and the stepwise experiment, and the Domain Corpus for the comparative experiment.

4.1.1 Common Corpus

This corpus has 1,000 positive and 1,000 negative sentences which were collected randomly from the Web by the author. We asked two Japanese native speakers (Experimenter A and B) to check these sentences and obtained an agreement of $\kappa = 0.92$ which indicates very high consistency (Table 4).

4.1.2 Domain Corpus

We also created a Domain Corpus, shown in Table 5. In this corpus, there are three benchmark datasets which have 200 positive and 200 negative opinion sentences respectively. The datasets are from the Electronic Products, Cars and Travels domains. The sentences were collected from reviews in "Kakaku.com†." We chose "Kakaku.com" because it is the largest Japanese Weblog specializing in product comparisons of consumer goods, including price and user opinions, etc. The website classifies the reviews as questions, positive reviews, negative reviews, rumors, sale information or "other". The sentences populating the datasets were collected from reviews that have been labeled as either positive or negative reviews. They were also checked by two native speakers, only sentences where both agreed were included in the data.

Table 4 Estimation results on common corpus by two experimenters.

		ExperimenterB				
		P	nonP	N	nonN	Total
Experi-menterA	P	924	12	0	0	936
	nonP	37	27	0	0	64
	N	0	0	953	16	969
	nonN	0	0	25	6	31
	Total	961	39	978	22	2000

ExperimenterA (P:936/N:969); ExperimenterB (P:961/N:978)
P: The number of positive sentences
nonP: The number of non-positive sentences
N: The number of negative sentences
nonN: The number of non-negative sentences

Table 5 Domain corpus.

Domain Name	PS	NS
Electronic Products	200	200
Cars	200	200
Travels	200	200

PS: The number of positive sentences
NS: The number of negative sentences

†http://www.kakaku.com/

4.2 Evaluation Setup

Our SO-PMI works on phrases. In the experimental evaluation we classify sentences. To classify a sentence as positive (P) or negative (N), the average SO of the phrases in the sentence is used according to Formula (11).

$$SO(sentence) = \frac{1}{n}\sum_{k=1}^{n}(SO(P_k)) \tag{11}$$

P : *phrase*

n : *the number of phrases in the sentence*

If the $SO(sentence)$ is greater than 0, the sentence is a positive sentence; otherwise it is a negative sentence.

As evaluation metrics, we measured our proposed method's performance by *accuracy*. *Accuracy* was measured as the number of sentences correctly classified as P/N sentences divided by the total number of P/N sentences in each benchmark dataset (Common Corpus: 1,000; Domain Corpus: 200). **PA** means *Positive Accuracy*, **NA** means *Negative Accuracy*, i.e. the accuracy on only positive or negative sentences respectively.

4.3 Balancing Factor Experiment

A good value for the balancing factor "α" was determined using the Common Corpus. The results are shown in Fig. 4. The dashed line in (a) and (b) indicates average accuracy (74%) on English Data from Turney's Study [9]. Turney did not evaluate positive and negative accuracy separately. The fully drawn line indicates the result after translating the original SO-PMI to Japanese (**PA**: 14.5%, **NA**: 94.3%). **PA** series (the line with triangle marks)/**NA** series (the line with circle marks) show the results for different values of "α" from 0 to 1.

Changing the α tends to be a tradeoff, lowering **PA** when **NA** is improved and vice versa. Therefore, we used the *Harmonic Mean* by the following formula to find a good choice of "α".

$$Harmonic\ Mean = \frac{2 \times PA \times NA}{PA + NA} \tag{12}$$

Figure 4 (c) shows **PA**, **NA** and *Harmonic Mean* curves for different values of "α". We selected "α=0.8" giving the highest *Harmonic Mean*, thus giving a good balance between **PA** (67.1%) and **NA** (61.6%).

4.4 Neutral Expression Detection Thresholds

This section presents the specific values (Formula (13)) of thresholds $\mathbf{t_a}$, $\mathbf{t_b}$ and $\mathbf{t_c}$ described in Neutral Expression Detection (3.3). These values were obtained using a small, hand-labeled corpus. The following values are used in the Stepwise Experiment (4.5) and the Comparative Experiment (4.6).

$$\mathbf{t_a}: 10000000;\ \mathbf{t_b}: 5 \tag{13}$$

$$\mathbf{t_c} = \begin{cases} 1000 & (min > 10000) \\ 100 & (1000 < min \leq 10000) \\ 10 & (100 < min \leq 1000) \\ 0 & (min \leq 100) \end{cases}$$

min : *minimum of* [*hits*(*phrase AND p_set*), *hits*(*phrase AND n_set*)]

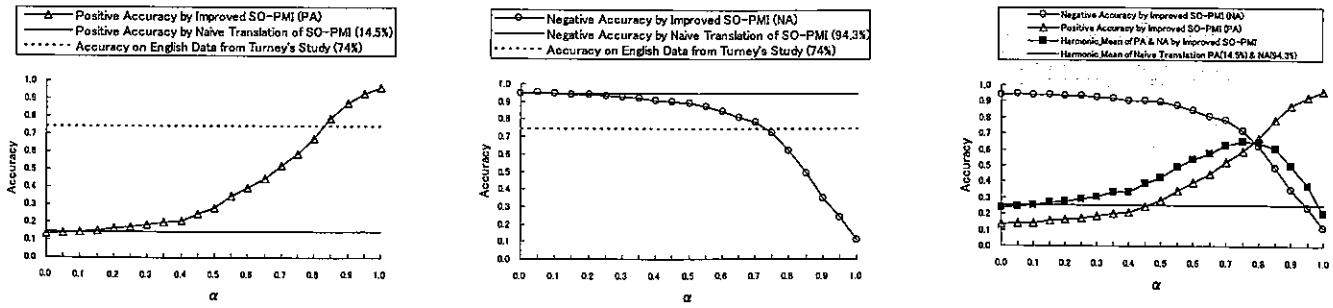
4.5 Stepwise Experiment

For evaluating the effects of the three improvements separately, we performed a stepwise experiment on the Common Corpus. The results are shown in Table 6. Test 1-4 are the tests for the naive translation of SO-PMI (NT-SO), adding the balancing factor (BF) and neutral expression de-

Table 6 Stepwise experiment on common corpus.

No.	Test Content	PA(%)	NA(%)	AA(%)
Test 1	NT-SO	14.5	94.3	54.4
Test 2	Test 1+BF	46.6	70.6	58.6
Test 3	Test 1+NED	16	93.2	54.6
Test 4	Test 1+BF+NED	49.4	75	62.2
Test 5	Improvement 1: RS	16.2	95	55.6
Test 6	Test 5+BF	70.1	60.2	65.2
Test 7	Test 5+NED	19.3	93.1	56.2
Test 8	Test 5+BF+NED	74.3	65	69.7

NT-SO: Naive Translation of SO-PMI for Japanese
RS: Improvement 1 - Reference Sets (3.1)
BF: Improvement 2 - Balancing Factor (3.2)
NED: Improvement 3 - Neutral Expression Detection (3.3)
AA: The average of PA and NA



(a) Positive Accuracy (PA)

(b) Negative Accuracy (NA)

(c) Harmonic Mean of PA and NA

Fig. 4 Experiment for balancing factor α.

tection (NED). Test 5-8 used the proposed reference sets, the BF and the NED step by step. The results show that the BF and the NED are also effective in the naive method, independent of reference words/sets. The effect of NED is dependent on the BF. After obtaining a balanced SO value, NED improved the accuracy by about 5%. As a whole, the Improved-SO (Test 8) gave more balanced results than the NT-SO (Test 4).

4.6 Comparative Experiment

The results of comparing a supervised system called Sentiment Analysis - demo (SA-Demo), the naive translation of the SO-PMI for Japanese (NT-SO) and our proposed improved SO-PMI (Improved-SO) on the Domain Corpus (4.1.2) are shown in Table 7.

SA-Demo[†]is an online sentiment analysis site which can present the number of positive/negative phrases in a given sentence [15]. We used this demo to classify sentences based on the number of positive/negative phrases. If the number of positive phrases is less than the number of negative phrases, the sentence is classified as a negative sentence. Otherwise, the sentence is classified as a positive sentence. The NT-SO approach is the naive translation of SO-PMI for Japanese: replacing the reference words "excellent" and "poor" with "いい (good)" and "悪い (bad)" in the SO formula (4.). The Improved-SO is our proposed approach using three improvements: Reference Sets (3.1), Balancing Factor (3.2) and Neutral Expression Detection (3.3).

In this experiment, we evaluated the classification accuracy of these three approaches on three domains (Electronic Products, Cars, and Travels) with 200 positive sentences and 200 negative sentences respectively. From the results (Table 7), we can conclude:

- These three approaches behave consistently on all three domains. The **PA** and **NA** for each domain were very similar.
- The **PA** of SA-Demo approach was higher than **NA**. All of the **PA**s exceeded 73%. On the other hand, none of the **NA**s were over 60%. SA-Demo had the highest **AA** (66.8%).
- As in the preliminary experiment in 2.2, the NT-SO approach produced slanted results. All of the **NA**s exceeded 94% and all of the **PA**s were under 11%. It had the lowest **AA** (52.3%).
- Our proposed SO-PMI approach (Improved-SO) achieved very balanced results: all of the **PA**s and **NA**s were from 62% to 68.5%. The **AA** was 65.7%, which was quite similar to the **AA** (66.8%) of the SA-Demo approach, despite our method being unsupervised and SA-Demo being a supervised method.

Table 7 Comparative experiment on domain corpus.

Domain Corpus		SA-Demo	NT-SO	Improved-SO
Electronic Products	PA(%)	73	9	68.5
	NA(%)	58	95	65.5
Cars	PA(%)	79.5	9	65
	NA(%)	52.5	95.5	62
Travels	PA(%)	78	11	65.5
	NA(%)	59.5	94.5	67.5
AA(%)		66.8	52.3	65.7

SA-Demo: Sentiment Analysis - demo
NT-SO: Naive Translation of SO-PMI for Japanese
Improved-SO: Our proposed SO-PMI using three improvements
AA: The average of all PA and NA

4.7 Discussion

In the three domain datasets the reputation expressions for different domains are very similar. For instance, "この仕様は非常に不満です (This specification is very unsatisfactory)" is an opinion from the Car domain, and "英英辞典が少々不満です (The British British dictionary is a little unsatisfactory)" is an opinion from Electronic Products domain. Since common reputation words were used in reference sets, we obtained results independent of any domain. In addition, since SA-Demo is based on a supervised approach, it can analyze explicit expressions (such as "最悪 (the worst)") precisely. However, for latent or undefined expressions (such as "使い物にならない (it is of no use)"), SA-Demo did not work well. In contrast, our proposed SO-PMI algorithm can perform well even in this case (SO (使い物にならない) = −0.157).

Another question for future work is the relationship of the accuracy of the estimation of semantic orientation at the level of individual phrases and the accuracy of sentence classification. Since the sentence classification is based on an average, it might be quite resistant to noise in the SO estimates of individual phrases. Another problem in this study is opinion phrase extraction. As mentioned in 3.3, adverbs that represent opinion, for example "ザラザラ (roughly)" or "しっかり (firm)", did not obtain any proper SO values because adverbs were not considered as opinion words.

As Turney mentioned in his paper [9], one limitation of SO-PMI is the time required to send queries to the search engine. Inspection of Formula (10) shows that it takes six queries to calculate the semantic orientation (SO) of a phrase. Since there is no need to recalculate $hits(p_set)$ and $hits(n_set)$ for every phrase, each phrase requires an average of slightly less than two queries. As a courtesy to Google search engine, we used a 0.5 second delay between queries. The domain datasets with 200 positive sentences and 200 negative sentences yielded on average 1,600 opinion phrases each, thus the total time required to process one domain dataset was roughly 1,600 seconds, about 27 minutes. This processing time is about 10 times faster than the time reported by Turney.

5. Application

Combining the proposed SO-PMI method and a Naive Bayes Classifier, we built a simple opinion mining system for analyzing Japanese Weblog reviews called OMS-J [16].

[†]http://www.cisr.iri.titech.ac.jp/~inui/cgi-bin/sa-demo/sa.cgi

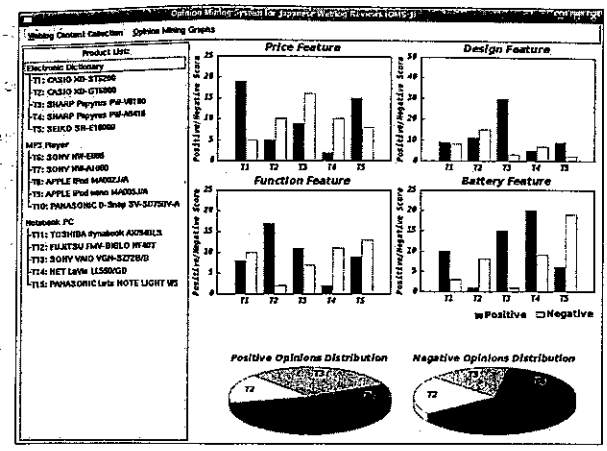


Fig. 5 OMS-J's GUI screenshot for one product category.

OMS-J is designed to provide an intuitive visual GUI of opinion mining graphs for a comparison of different products of the same type to help a user make a quick purchase decision.

There are two classifications in this system: Feature classification and P/N (positive/negative) classification. Feature means a product's attribute, i.e. price, design, function or battery feature. Since it is easy to create a feature corpus, this classification is performed by a supervised approach, a Naive Bayes Classifier. P/N classification classifies expressions into positive or negative meaning using the unsupervised approach improved SO-PMI. Thus, the system can provide graphs (Fig. 5) that show the distribution of positive and negative opinions of each feature based on the results of these two classifications.

6. Related Work

6.1 Research Using SO-PMI

Since Turney [9], [17] proposed the SO-PMI algorithm and showed the effectiveness when used on English, the SO-PMI algorithm has attracted more and more interest, since it uses the Web as a corpus to estimate the semantic orientation (SO) of a phrase by measuring the co-occurrences with known positive or negative words. This is considered an effective and important method for opinion analysis in the current information flood age. There is much research that uses or expands this algorithm. Popescu and Etzioni [2] introduced an unsupervised information-extraction system (OPINE) using SO-PMI which mines reviews in order to build a model of important product features, their evaluation by reviewers and their relative quality across products. Chaovalit's study [3] gave a comparison between a supervised approach (N-gram Classifiers) and an unsupervised approach (SO-PMI) on movie opinion mining. Yu and Hatzivassiloglou [4] proposed several models for an opinion question answering system: distinguishing between opinions and facts at the document level, and classifying opinion sentences as positive, negative or neutral using SO-PMI. Taboada et al [18] presented an expanded SO-PMI method by giving more prominence to certain parts of the text where they believe most subjective content is concentrated.

All of this research deals with English opinion data. To the best of our knowledge, research that attempts to adapt SO-PMI to Japanese has not yet been reported. In this paper, we showed that our proposed method not only adapted the SO-PMI for Japanese, but also improved it to analyze Japanese opinions more effectively.

6.2 Other Work

Using other approaches to do opinion mining, Pang et al. [5] adopted a more direct approach. They achieved up to 83% precision using supervised learning with words and n-grams as features to predict orientation at the document level. Pang et al. [6] used minimum cuts to improve the classification of a sentence into either "objective" or "subjective", with the assumption that sentences close to each other tend to have the same class. Dave et al. [7] categorized positive versus negative movie reviews using Support Vector Machines on various types of semantic features based on substitutions and proximity, and achieved an accuracy of 88.9% on data from Amazon and cnn.net. Gamon [19] used Support Vector Machines with large lexical and syntactic feature sets to show that the addition of deep linguistic analysis features to a set of surface level word n-gram features contributes consistently to classification accuracy. Liu et al. [20] proposed a system for analyzing and comparing consumer opinions of competing products. They also presented a language pattern based method for extraction of product features. Wilson et al. [21] investigated how context influences sentiment polarity at the phrase level in a corpus, beginning with a predefined list of words with polarities. All this research is based on traditional supervised approaches. In contrast, our study relies on an unsupervised approach.

For Japanese opinion mining, Kobayashi et al. [13] defined the main portions of an opinion as <object, attribute, opinion>. They proposed a technique for efficiently building an object name dictionary, an attribute expression dictionary, and an opinion word dictionary for any object domain. Suzuki et al. [22] presented a semi-supervised method by combining a Naive Bayes classifier and the EM algorithm to classify the Weblog reputation information into positive, negative or neutral categories. As for English, the Japanese opinion mining is generally done using supervised methods, unlike our unsupervised approach.

6.3 SA-Demo

Takamura et al. [15] proposed a supervised learning approach which introduced latent variables into the models for semantic orientations of phrases as well as classification methods based on the models. They achieved nearly 82% classification accuracy, and provide an online Sentiment Analysis demo system using their proposed approach. This system uses a hand-labeled training corpus and can ob-

tain precise classification results. It is a well known approach in the field of Japanese sentiment analysis. In the comparative experiments of this paper, we compared our unsupervised method of positive and negative classification accuracy with this system. The results showed that this SA-Demo system gave *Positive Accuracies* of near 80%, but the *Negative Accuracies* were all under 60%. And our proposed method achieved similar *Average Accuracy* with it.

7. Conclusions

This study proposed an improved unsupervised approach (SO-PMI) for Japanese Weblog Opinion Mining. Some parts of Turney's original approach, such as the *NEAR* operator, do not work for Japanese, thus some modifications must be done. We introduced three improvements based on the characteristics of Japanese and the results of related work. The experiment results (*Average Accuracy*: 65.7%) showed that our proposal achieved a considerably improved performance, compared to naively translating the SO-PMI. They also showed that the improved unsupervised approach can produce results comparable to a supervised approach. Moreover, this not only shows that the SO-PMI algorithm can also work for other languages besides English, but also gives useful hints for other Asian languages which have similar characteristics (no space between words etc).

It would be expected that the three proposed improvements would also work effectively for other reference words or languages. In the future, we will try out different choices of words for the sets of positive and negative reference words, and enhance the analysis accuracy of individual phrases. We also plan to evaluate our proposal on other languages.

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